

Manuel Bendaña

PhD Student

Centro Singular de Investigación en Tecnoloxías Intelixentes (CiTIUS) · Universidade de Santiago de Compostela

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Summary

PhD student in Computer Vision at the Universidade de Santiago de Compostela and researcher at CiTIUS, supervised by Prof. Manuel Mucientes and Prof. Víctor M. Brea. My research focuses on deep learning for **Multiple Object Tracking (MOT)**, together with object detection and segmentation. Holder of a highly competitive FPU predoctoral grant from the Spanish Ministry of Universities.

Education

2024 – Present	PhD in Information Technology Research , <i>Universidade de Santiago de Compostela</i> Thesis: Visual Multi Object Tracking with frequent occlusions. Advisors: Prof. Manuel Mucientes and Prof. Víctor M. Brea.
2022 – 2024	MSc in Computer Vision , <i>Universidade de Santiago de Compostela, Universidade da Coruña, Universidade de Vigo and Universidade do Porto</i>
2018 – 2022	BSc in Computer Science , <i>Universidade de Santiago de Compostela</i>

Research experience

Jan 2024 – Present	Predoctoral Researcher , <i>CiTIUS — Centro Singular de Investigación en Tecnoloxías Intelixentes</i> FPU-funded PhD research on ‘Visual Multi Object Tracking with frequent occlusions’.
Jan 2023 – Oct 2023	Junior Researcher , <i>CiTIUS — Centro Singular de Investigación en Tecnoloxías Intelixentes</i> Research on Object Detection and Visual Object Tracking.
Mar 2022 – May 2022	Research Internship , <i>CiTIUS — Centro Singular de Investigación en Tecnoloxías Intelixentes</i> Curricular Internship (BSc Degree in Computer Science): ‘Development of video analytics systems using detection and tracking techniques’.
2021 – 2022	Research Intern , <i>CiTIUS — Centro Singular de Investigación en Tecnoloxías Intelixentes</i> Research on ‘Object detection using few-shot techniques’, funded by a Collaboration Grant (Spanish Ministry of Education).
Jul 2021	Research Intern , <i>CiTIUS — Centro Singular de Investigación en Tecnoloxías Intelixentes</i> Summer internship on ‘Object detection in video with few examples’.

Publications

Journal articles

- [J1] **Segmentation-guided association refinement in multiple object tracking**
M. Bendaña, V. M. Brea, M. Mucientes. *Neurocomputing*, vol. 708, pp. 135176, 2027 (Available online).
doi:10.1016/j.neucom.2026.135176

Conference papers

- [C1] **Live Demonstration: Transformer-Based Visual Object Detection and Tracking on MPSoC**
M. Romero-Romero, **M. Bendaña**, P. Gil-Pérez, D. Cores, F. Pardo, V. M. Brea, M. Mucientes. *2026 IEEE International Symposium on Circuits and Systems. Demonstration*. doi:10.1109/ISCAS66217.2026.11562030
- [C2] **Enhancing Multi-object Tracking with Segmentation Masks: A Solution for Lost Object Recovery**
M. Bendaña, L. Vaquero, V. M. Brea, M. Mucientes. *12th Iberian Conference on Pattern Recognition and Image Analysis, 2025. Oral presentation*. doi:10.1007/978-3-031-99565-1_6

Research projects

2024 – Present	AZOR — Search, Location and Rescue of People on Land, and Fire Fighting and Prevention , <i>Researcher. Civil UAVs Initiative: Fase B, Xunta de Galicia (Galicia, Spain) - Axencia Galega de Innovación (GAIN)</i>
2024 – 2027	REOPEN: Leveraging Artificial Intelligence for Robust Predictive Monitoring in Process Mining , <i>Researcher. Ministerio de Ciencia e Innovación (Spain)</i>

Awards & grants

Awards

2025	MSc in Computer Vision: Best Thesis Award, Extraordinary Award and Top Academic Record , <i>Escola Técnica Superior de Enxeñaría, Universidade de Santiago de Compostela</i>
2023	BSc in Computer Science: Extraordinary Award and Top Academic Record , <i>Escola Técnica Superior de Enxeñaría, Universidade de Santiago de Compostela</i>
2023	Extraordinary Graduation Award (BSc in Computer Science) , <i>Xunta de Galicia, Spain</i>
2020 & 2021	Academic Excellence Awards (academic years 2018/2019 and 2019/2020) , <i>Xunta de Galicia, Spain</i>

Grants

2023	Formación del Profesorado Universitario (FPU) Predoctoral Research Grant , <i>Ministry of Universities, Spain – Highly competitive national call</i>
2021	Collaboration Grant , <i>Ministry of Education, Spain</i>

Teaching

2026–2027 2024–2025	Neural Networks and Deep Learning (BSc in Artificial Intelligence) , <i>Universidade de Santiago de Compostela. Teaching assistant · lab sessions · 55 h</i>
2026–2027 2024–2025	Automata Theory and Formal Languages (BSc in Computer Science) , <i>Universidade de Santiago de Compostela. Teaching assistant · lab sessions · 65 h</i>
2025–2026	Automata and Formal Languages (BSc in Artificial Intelligence) , <i>Universidade de Santiago de Compostela. Teaching assistant · lab sessions · 60 h</i>

Talks

Oct 2025	Segmentation-Guided Association Refinement in Multiple Object Tracking , <i>Talk. Doctoral Meeting. CiTIUS, Santiago de Compostela, Spain</i>
Jul 2025	Enhancing Multi-Object Tracking with Segmentation Masks: A Solution for Lost Object Recovery , <i>Talk. IbPRIA 2025 — 12th Iberian Conference on Pattern Recognition and Image Analysis. Coimbra, Portugal</i>

Skills

Programming	Python · C/C++ · Java · Bash
Deep learning	PyTorch · TensorFlow · scikit-learn · OpenCV · TrackEval · detectron2 · SAM 2
Tools	Git · Docker · Linux · LaTeX
Languages	Spanish (native) · Galician (native) · English (B1, certified; working language for research papers and conference talks)